

Clarkson Institute Summer School:

Examples Sheet 1 - Groups: definitions, examples, and isomorphism

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Definition of a group

1. Decide which of the following sets, with the given binary operations, constitute groups. In each case, verify the group axioms, or determine which of them do not hold.
 - (a) the integers $\mathbb{Z}$, with the operation $a * b = a - b$;
 - (b) the complex numbers $\mathbb{C}$, with the operation $a * b = ab / (a + b)$;
 - (c) the set of odd integers, with the operation $a * b = a + b + 1$.

Solution:

- (a) This is not associative, since $a * (b * c) = a - (b - c) = a - b + c$, but $(a * b) * c = (a - b) - c = a - b - c$. Hence this cannot be a group.

Additionally, there is no identity, since $a * b = a$ for all b requires that $a - b = a$ so that $b = 0$. So no such a can satisfy this. In particular, this precludes the existence of inverses, because there is no identity in the first place.

- (b) This is not closed, since $a * b = ab / (a + b)$ does not exist in the complex numbers when $a + b = 0$, e.g. when $a = b = 0$.

However, the operation *is* associative, since:

$$a * (b * c) = a * \frac{bc}{b+c} = \frac{a(bc)/(b+c)}{a+bc/(b+c)} = \frac{abc}{ab+ac+bc}.$$

Similarly, this is also equal to $(a * b) * c$.

An identity e would need to satisfy $a * e = a$ for all a . This would imply $ae = a(a + e)$ for all a , which requires $a = 0$; this is a contradiction. Thus there is no identity, and there are no inverses.

- (c) This is a group. We check each of the axioms in turn:

(G1) *Closure*. The operation is closed, since if a, b are odd, then $a + b$ is even, and hence $a + b + 1$ is odd.

(G2) *Associativity*. The operation is associative, since:

$$a * (b * c) = a * (b + c + 1) = a + (b + c + 1) + 1 = a + b + c + 2,$$

$$(a * b) * c = (a + b + 1) * c = (a + b + 1) + c + 1 = a + b + c + 2.$$

(G3) *Identity*. The identity is -1 , since $a * (-1) = a + (-1) + 1 = a$, and $(-1) * a = (-1) + a + 1 = a$.

(G4) *Inverses*. The inverse of a is $a^{-1} = -a - 2$, since:

$$a * a^{-1} = a + (-a - 2) + 1 = -1, \quad a^{-1} * a = (-a - 2) + a + 1 = -1.$$

Examples of groups from arithmetic

2. Identify the elements and construct the group table for $\mathbb{Z}_8^*$. What feature of the group table implies that the group is Abelian? Give the inverses of all the elements.

Solution: The elements are $\{1, 3, 5, 7\}$. The group table is:

$\cdot$	1	3	5	7
1	1	3	5	7
3	3	1	7	5
5	5	7	1	3
7	7	5	3	1

The fact that the group table is reflectionally symmetric about the leading diagonal implies that the group is Abelian, since it tells us that $g \cdot h = h \cdot g$ for all g, h in the group. The inverses of the elements are $1^{-1} = 1, 3^{-1} = 3, 5^{-1} = 5, 7^{-1} = 7$; that is, all elements are self-inverse.

3. Find the multiplicative inverse q^{-1} of the quaternion $q = 1 + i + j + k$. Verify directly that we have $qq^{-1} = 1 = q^{-1}q$.

Solution: Motivated by the way in which we find inverses of complex numbers, we conjecture that the inverse is:

$$\frac{1 - i - j - k}{4}.$$

Checking this, we have:

$$\begin{aligned} qq^{-1} &= \frac{1}{4}(1 + i + j + k)(1 - i - j - k) \\ &= \frac{1}{4}(1 - i - j - k + i + 1 - k + j + j + k + 1 - i + k - j + i + 1) \\ &= 1. \end{aligned}$$

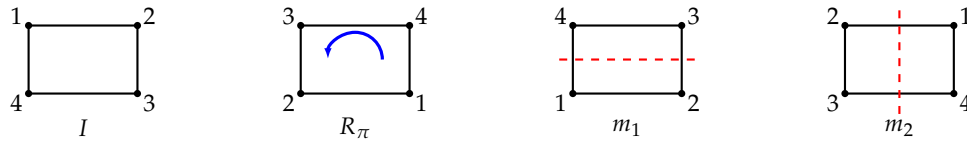
Similarly, we have:

$$q^{-1}q = \frac{1}{4}(1 - i - j - k)(1 + i + j + k) = \dots = 1.$$

Examples of groups from geometry

4. Construct the group table for the set of symmetries of a non-square rectangle.

Solution: Let the symmetries be $\{I, R_\pi, m_1, m_2\}$, where I is doing nothing, R_π is a rotation by 180° , m_1 is a reflection in the horizontal axis, and m_2 is a reflection in the vertical axis. The symmetries are shown below, with the vertices labelled 1 to 4 to better keep track of how the transformations affect the rectangle.



The group table is:

$\circ$	I	R_π	m_1	m_2
I	I	R_π	m_1	m_2
R_π	R_π	I	m_2	m_1
m_1	m_1	m_2	I	R_π
m_2	m_2	m_1	R_π	I

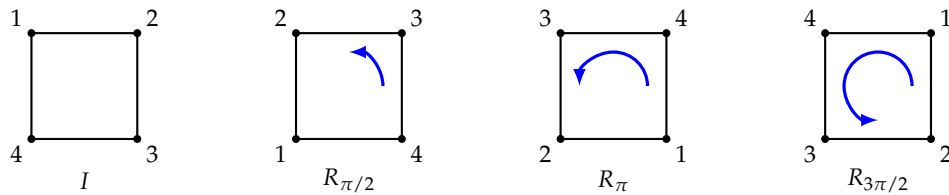
5. Construct the group table for D_4 . What feature of the group table implies that the group is non-Abelian? Give the inverses of all the elements.

Solution: The elements of D_4 are given by:

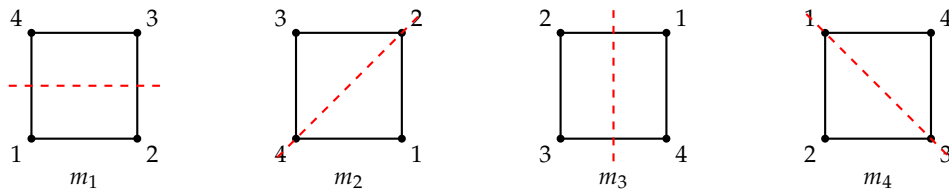
$$\{I, R_{\pi/2}, R_\pi, R_{3\pi/2}, m_1, m_2, m_3, m_4\},$$

with the relevant symmetries shown in the diagram below.

Rotations



Reflections



To quickly produce the group table, note that the multiplication of the rotations is obvious, given by $R_\theta R_\phi = R_{\theta+\phi}$. Also note that by the discussion in the lecture, we have:

$$R_\theta m_i = m_i R_{2\pi-\theta};$$

that is, performing a reflection followed by a rotation is equivalent to performing the same rotation *in the opposite sense*, followed by the reflection. We also have $m_i^2 = I$ for each of $i = 1, 2, 3, 4$.

The remaining products we need to work out are $R_\theta m_i$ for each of the possible θ, i , and $m_i m_j$ for $i \neq j$. Evaluating them similarly to the lecture, we find that the group table is:

$\circ$	I	$R_{\pi/2}$	R_π	$R_{3\pi/2}$	m_1	m_2	m_3	m_4
I	I	$R_{\pi/2}$	R_π	$R_{3\pi/2}$	m_1	m_2	m_3	m_4
$R_{\pi/2}$	$R_{\pi/2}$	R_π	$R_{3\pi/2}$	I	m_2	m_3	m_4	m_1
R_π	R_π	$R_{3\pi/2}$	I	$R_{\pi/2}$	m_3	m_4	m_1	m_2
$R_{3\pi/2}$	$R_{3\pi/2}$	I	$R_{\pi/2}$	R_π	m_4	m_1	m_2	m_3
m_1	m_1	m_4	m_3	m_2	I	$R_{3\pi/2}$	R_π	$R_{\pi/2}$
m_2	m_2	m_1	m_4	m_3	$R_{\pi/2}$	I	$R_{3\pi/2}$	R_π
m_3	m_3	m_2	m_1	m_4	R_π	$R_{\pi/2}$	I	$R_{3\pi/2}$
m_4	m_4	m_3	m_2	m_1	$R_{3\pi/2}$	R_π	$R_{\pi/2}$	I

6. In theoretical particle physics, the Universe possesses three basic discrete symmetries: *parity inversion*, P , which sends a spatial point (x, y, z) to $(-x, -y, -z)$; *time reversal*, T , which reverses the flow of time; and *charge conjugation*, C , which changes the sign of the charge of all charged particles in the Universe. Together with the identity symmetry, they form a group of symmetries $G = \{I, C, P, T\}$. The *CPT theorem* from physics says that:

$$CPT = I.$$

Using this theorem, construct the group table for G , and show that the group is Abelian. [Hint: eliminate one of P, C, T in terms of the other two symmetries.]

Solution: From the relation $CPT = I$, we see that $CP = T^{-1}$. But $T^2 = I$ (reversing the direction of time twice must do nothing overall), hence we have $T^{-1} = T$. This means we can eliminate T in favour of $T = CP$.

Therefore, the group elements are $\{I, C, P, CP\}$. Taking the square of CP , observe that:

$$I = T^2 = (CP)^2 = (CP)(CP).$$

Multiplying on the left by C and the right by P , we have $CP = PC$, since $C^2 = P^2 = I$, similarly to T^2 (this is because changing the sign of all charges in the Universe twice does nothing overall, and performing parity inversion twice does nothing overall). Hence C, P commute. Using this, together with the fact that $C^2 = P^2 = I$, we can easily construct the group table as:

$\circ$	I	C	P	T
I	I	C	P	T
C	C	I	T	P
P	P	T	I	C
T	T	P	C	I

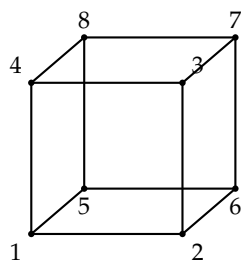
Here, we use identities like $CT = C(CP) = P$, and $PT = P(CP) = CP^2 = C$.

7. (a) Determine the orders of the chiral cubic group (the set of rotational symmetries of a cube) and the full cubic group (the set of rotational and reflectional symmetries of a cube). Are these groups Abelian or non-Abelian? Justify your answers.
- (b) Repeat part (a) for the corresponding octahedral groups. What do you notice about your answers, and why?

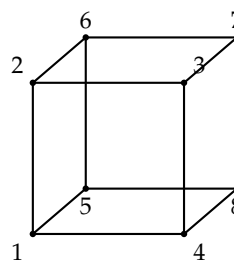
Solution:

- (a) Label the vertices of the cube 1 to 8 (for example, as in the figure on the left below). Any symmetry must take a vertex to a vertex, so there are eight possible positions for the 1-vertex. Once we have mapped the 1-vertex to another vertex, we must choose a connected vertex for the new position of the 2-vertex. There are three possible choices for this. Once we have chosen both the 1 and 2 vertices, the remaining vertices are determined because they must have the same handedness as the original cube (if we perform only rotations, and no reflections). As a result, there are $8 \cdot 3 = 24$ rotational symmetries of the cube.

The handedness is shown in the below figure: for example, in a 'right-handed configuration' shown on the left, if 1 is initially connected to 2, 4, 5 as shown, then you can put your right hand at the 1-vertex, your index finger in the direction of the 2-vertex, and your middle finger in the direction of the 5-vertex, and your thumb in the direction of the 3-vertex. A rotation cannot 'reflect a right-hand into a left-hand', hence the handedness of the cube cannot change under a rotation.



'Right-handed'

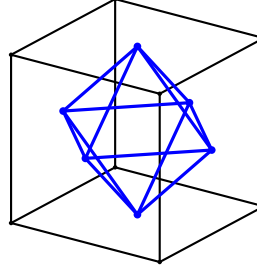


'Left-handed'

If we now allow for reflectional symmetries, the 3-vertex can be connected to 1 in two possible places, because of the freedom to change the handedness of the cube. This gives a total of $8 \cdot 3 \cdot 2 = 48$ rotational and reflectional symmetries of the cube.

In both cases, the groups are non-Abelian; picking any two sensible rotations will reveal this.

- (b) The exact same argument for counting carries through to the octahedron, so the orders are identical. In fact, more strongly, the cube and octahedron have exactly the same symmetries. This is because they are *dual polyhedra*; we can inscribe an octahedron inside a cube as shown in the figure below, by joining the centres of all the cube faces.



Any symmetry of the cube must therefore also be a symmetry of the octahedron, and vice-versa.

Examples of groups from matrix algebra

8. Show that the powers of the matrix:

$$P = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}$$

form a group under matrix multiplication. Construct the group table.

Solution: The powers of the matrix can be computed as:

$$P^2 = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}, \quad P^3 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad P^4 = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix},$$

$$P^5 = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}, \quad P^6 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I.$$

Therefore the distinct powers are

$$\{I, P, P^2, P^3, P^4, P^5\},$$

since $P^6 = I$. The group multiplication law is

$$P^i P^j = P^{i+j},$$

where the exponent is taken modulo 6. The group table is therefore:

$\cdot$	I	P	P^2	P^3	P^4	P^5
I	I	P	P^2	P^3	P^4	P^5
P	P	P^2	P^3	P^4	P^5	I
P^2	P^2	P^3	P^4	P^5	I	P
P^3	P^3	P^4	P^5	I	P	P^2
P^4	P^4	P^5	I	P	P^2	P^3
P^5	P^5	I	P	P^2	P^3	P^4

9. Show that the following set of matrices forms a group under matrix multiplication, and construct their group table:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad A = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.$$

Solution: Working out each of the possible products, we have:

$$A^2 = I, \quad B^2 = I, \quad C^2 = I,$$

$$AB = C, \quad BA = C,$$

$$AC = B, \quad CA = B,$$

$$BC = A, \quad CB = A.$$

Thus the set is closed under matrix multiplication. Since matrix multiplication is associative, and I is the identity matrix, it follows that these matrices form a group (with every element self-inverse). The group table is:

$\cdot$	I	A	B	C
I	I	A	B	C
A	A	I	C	B
B	B	C	I	A
C	C	B	A	I

10. Is it true that a set of matrices forms a group under multiplication only if all matrices in the set are invertible? Provide a proof or counterexample as appropriate.

Solution: This is false, e.g.

$$G = \left\{ \begin{pmatrix} x & 0 \\ 0 & 0 \end{pmatrix} : x \in \mathbb{R}^* \right\}$$

is a set of 2×2 matrices, all of which are non-invertible, that form a group under matrix multiplication.

Isomorphic groups

11. Look at the group tables you constructed in Questions 2, 4, 6, 8, and 9. In each case, identify a cyclic or dihedral group from this chapter to which the group is isomorphic.

Solution: Considering each group in turn:

- In Question 2, the group is isomorphic to D_2 .
- In Question 4, the group is also isomorphic to D_2 .
- In Question 6, the group is also isomorphic to D_2 .
- In Question 8, the group is isomorphic to C_6 .
- In Question 9, the group is isomorphic to D_2 .

12. Show that if $f : G_1 \rightarrow G_2$ is an isomorphism, then $f(g^{-1}) = f(g)^{-1}$ for all group elements $g \in G_1$. Deduce that C_4 and D_2 are not isomorphic groups.

Solution: From the lectures, we know that $f(e_1) = e_2$, where e_1, e_2 are the identities of G_1, G_2 respectively. Given any group element $g \in G_1$, we then have:

$$e_2 = f(e_1) = f(g * g^{-1}) = f(g)f(g^{-1}),$$

so that $f(g)^{-1} = f(g^{-1})$, as required.

This result shows that if g, g^{-1} are inverses in the group G_1 , then $f(g), f(g^{-1})$ must be inverses in the group G_2 . In particular, a self-inverse element must be mapped to a self-inverse element. In C_4 , there are two self-inverse elements, I, R_π . On the other hand, in D_2 there are four self-inverse elements, I, R_π, m_1, m_2 . Therefore the two groups cannot be isomorphic.

13. Is Q_8 isomorphic to a cyclic or dihedral group?

Solution: If Q_8 were isomorphic to a cyclic group, it could only be isomorphic to C_8 , by considering sizes. Similarly if Q_8 were isomorphic to a dihedral group, it could only be isomorphic to D_4 , again by considering sizes.

Note that C_8 is Abelian, but Q_8 is non-Abelian, so they cannot be isomorphic.

The self-inverse elements in Q_8 are 1 and -1 only. On the other hand, in D_4 there are five self-inverse elements, I, m_1, m_2, m_3, m_4 , consisting of the identity symmetry and four reflections. Therefore, Q_8 cannot be isomorphic to D_4 by the result of the previous question.